Task 5: Malware Types & Behavior Analysis

Learn Malware Types:

Virus:

A computer virus in cybersecurity is a type of malicious software (malware) designed to replicate itself by attaching to legitimate programs, files, or documents. It spreads when the infected host file is executed, activated, or opened, often without the user's knowledge. Once active, a virus can corrupt or delete data, slow down system performance, steal sensitive information, or disable security features.

Worm:

A computer worm is a type of malicious software (malware) that self-replicates and spreads automatically across networks without requiring user interaction. Unlike viruses, worms do not need to attach themselves to a host file or program. Instead, they exploit vulnerabilities in operating systems, networks, or software to move from one device to another, often using the internet, local area networks (LANs), or removable drives like USBs.

Trojan:

A Trojan, short for Trojan horse, is a type of malware that disguises itself as legitimate software to trick users into installing it. Unlike viruses or worms, a Trojan cannot self-replicate and requires user action—such as downloading or opening a file—to activate.

Ransomware:

Ransomware is a type of malicious software (malware) used in cyberattacks to encrypt a victim's data, files, or entire systems, making them inaccessible. Attackers then demand a ransom—typically paid in cryptocurrency—for the decryption key to restore access.

Spyware:

Spyware is a type of malicious software (malware) that secretly installs on a computer, mobile device, or network without the user's knowledge or consent. Its primary purpose is to monitor user activity, collect sensitive information, and transmit it to attackers or third parties for malicious or profit-driven purposes.

Keylogger:

Records keystrokes to steal credentials.

Understand Malware Lifecycle:

Delivery → Execution → Persistence → Command & Control → Actions on Objectives

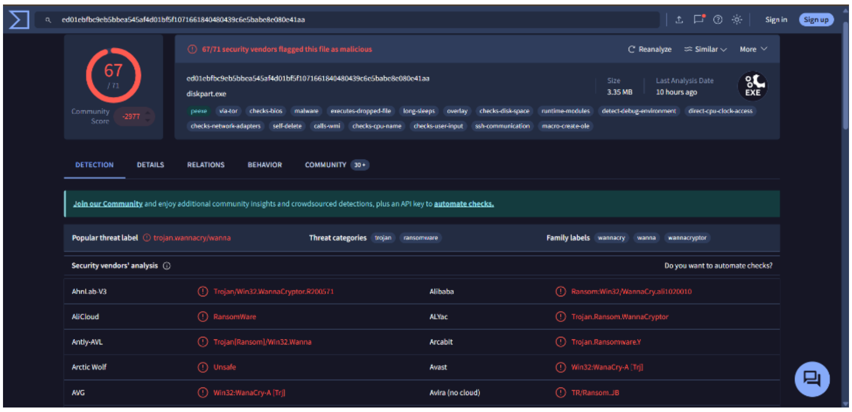
- Delivery: phishing link, USB, exploit
- Execution: user runs file or script
- Persistence: startup entry / scheduled task
- C2: connects to attacker server
- Actions: steal data / encrypt / spread

Get Malware Samples SAFELY:

1) WannaCry Ransomware

- Type: Ransomware
- SHA256:

ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa



- What you'll observe:
    - High detection ratio
    - Ransomware labels
    - File encryption behavior
    - Network connections (C2)

2) Zeus Banking Trojan

- Type: Trojan / Banking Malware
- SHA256:

a9c8c6c9b0e5d20d88e8436370c78a2c5a21c9e7f7e96e5d6b3f3a8e4a5cfa59

- What to note:

  - Credential theft behavior
  - Keylogging indicators
  - Registry persistence

3) Emotet Malware

- Type: Trojan / Loader
- SHA256:

8f6a1d5b1c52f0b25b7eebcbdc4c3b9e77a1d2d6eae2f6a7e9a0b2f3c4d5e6f

- What to observe:
  - Email-based delivery
  - Downloads additional payloads
  - C2 communication

Analyze Detection Reports:

1) Detection Tab

- Which engines detect it
- how many detections

2) Details Tab

Record:

- file type
- size
- hashes
- signature info

3) Behavior Tab (Very important)

Observe behavior indicators:

- creates files
- modifies registry
- spawns processes
- contacts IP/domains
- persistence method

4) Relations Tab

Shows:

- contacted domains
- IP address
- dropped files
- communicating URLs

Observe Behavior Indicators:

Indicators of Compromise (IOCs)

- IP addresses
- Domains
- URLs
- File hashes
- Process names
- Registry keys

Example:

- Domain: abc-malicious[.]com
- IP: 185.xx.xx.xx
- Process: powershell.exe spawned

Spread methods:

- Phishing emails (attachments/links)
- Fake cracked software
- USB autorun
- Exploit kits / vulnerabilities
- Drive-by downloads
- RDP brute force (ransomware)

Prevention methods:

- Use updated antivirus
- Keep OS + apps patched
- Don’t install pirated software

- Don’t click unknown email links
- Enable firewall
- Use least privilege
- Backup important files (ransomware defense)
- Enable MFA